NYC Subway Equity Analysis

Does NYC's subway serve all neighborhoods equally?

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496	29	4	1,540
Stations	Lines	Years	Observations

NYC Subway Equity Analysis - The Core Question

Does New York City's subway provide equal service quality across all neighborhoods?

An observational, data-driven equity study examining service frequency, delay rates, and ridership patterns by income level

Service Frequency

Do lower-income stations receive fewer scheduled peak-hour trips?

Peak-hour analysis: AM peak (GTFS)

Data available

Delay Rates

Do lines serving lower-income areas suffer more frequent delays?

Monthly delay incidents: MTA data

Data available

Ridership Recovery

Did income-based gaps widen or narrow post-2020 pandemic?

Weekday vs weekend ridership

Data available

Note: This is an **observational, data-driven equity study** — not causal inference. We analyze correlations between service quality and neighborhood demographics without establishing causation. Observational Study

Data Sources & Methodology

Four Independent Data Sources

Validated

Comprehensive data integration from multiple authoritative sources

MTA Operational Data

Ridership & delays data from 2020–2024, including service disruptions and incident reports

NY.gov open data portal

GTFS Static Feed

Station locations, schedules, and service frequencies for all subway lines

MTA API

U.S. Census ACS 2022

Tract-level income and demographic data for neighborhood analysis

Census Bureau

Census Geocoder

Maps stations to Census tracts for spatial analysis

Geocoding API

Key Contribution

Station-to-neighborhood linkage enabling equity analysis impossible from any single dataset. This crosswalk is the core analytical innovation.

Analysis Pipeline

1. **Station → Tract Crosswalk**
   Map each station to its Census tract via geocoder
   Completed
2. **Feature Engineering**
   Peak-hour trips, delay rates, ridership ratios
   Completed
3. **Demographics Merge**
   Attach income quartile & minority-majority flag
   Completed
4. **Statistical Testing**
   Partial correlation, Spearman, Mann-Whitney, Welch t
   Completed

Presentation Agenda

A comprehensive analysis of transit equity across NYC neighborhoods

1. **Motivation & Core Question**
   Understanding the equity gap in NYC subway service quality

2. **Data Sources & Methodology**
   MTA data, Census integration, and analysis pipeline

3. **Dataset Overview**
   496 stations, 1,540 observations, 29 lines

4. **Equity Framework & Hypotheses**
   H1-H4: Service frequency, delays, ridership patterns

5. **Statistical Results**
   Hypothesis testing with detailed findings

6. **Case Study Analysis**
   Rockaway Park vs 42nd St Shuttle comparison

Key Statistics

Total Stations Analyzed	496
Station-Year Observations	1,540
Subway Lines Covered	29

2 of 4 hypotheses supported

Data from MTA, Census Bureau, NY Open Data

NYC Subway Equity Analysis - Study Scope & Definitions

Study Scope

Observational Study

Unit of Analysis

Station-year observations

Delays aggregated at line level

Geography

NYC subway stations

Mapped to Census tracts

Timeframe

2020–2024

Pandemic disruption included

Purpose

Observational equity analysis

Not causal inference

Data Integration

Stations	Lines	Years
496	29	5

Data completeness: 92%

Key Definitions

5 definitions

Minority-majority station

Station mapped to tract with >50% residents from minority racial/ethnic groups

Income quartiles

Tract median income split into Q1–Q4 using ACS 2022 data

Peak-hour trips

Scheduled departures in AM peak (GTFS static schedule)

Delay incidents

Monthly/weekly service delay counts reported by MTA (line level)

Weekday–Weekend ridership gap

Relative difference in ridership between weekdays and weekends at station level

Executive Summary

Statistical analysis of NYC subway equity patterns

Key Findings Overview

4 Hypotheses Tested

Network-wide Signals

H1 & H2 Results

Service Frequency vs Income

$r = 0.039$, $p = 0.355$

95% CI: [-0.000, +0.074]

Not Supported H1

Delay Rate vs Line Income

$\rho = 0.198$, $p = 0.416$

95% CI: [-0.284, +0.625]

Not Supported H2

Clear Equity Disparities

H3 & H4 Results

Weekday vs Weekend Gap

$U = 101{,}416$, $p = 3.2 \times 10^{-8}$

95% CI: [0.087, 0.275]

Supported H3

Shuttle Delay Comparison

$t = 17.0$, $p = 1.8 \times 10^{-26}$

CI: [30.7, 38.6] incidents/month

Supported H4

Key Implications

Policy Impact

Transit as Work Tool

Lower-income riders depend more on transit

High Impact

Localized Disparities

Rockaway Shuttle shows 34 more delays/month

Critical

Strategic Implications

Equity is Not Universal

Income alone doesn't predict service quality at network scale

Commuter Patterns

Lower-income riders show stronger weekday dependence

Localized Disparities

Specific routes carry measurable service burden

Dataset at a Glance - NYC Subway Equity Analysis

Key Dataset Statistics

Data Validated

Unique Subway Stations
Total stations analyzed

496

stations across NYC

100% coverage

29 lines covered — Active

Station-Year Observations
Total data points

1,540

observations (2020-2024)

4 years of data

308 avg per year — Complete

Minority-Majority Stations
Racial demographic data

286/496

stations (58%)

58% of network

58% minority-majority — Analyzed

Years of Data
Time period covered

4

years (2020-2024)

Complete coverage

2020-24 period — Full

Income Range
Tract median income

$11K-$250K

range of median income

2022 ACS data

$79K median — Updated

Data Quality
Completeness & accuracy

92%

data completeness

Validated

0 missing — Verified

Data Source Information

Data sourced from NY.gov Open Data Portal, MTA API, and US Census Bureau 2022 ACS. All data accessed programmatically with no manual downloads required.

Automated

Equity Overview — Minority vs Majority Neighborhoods

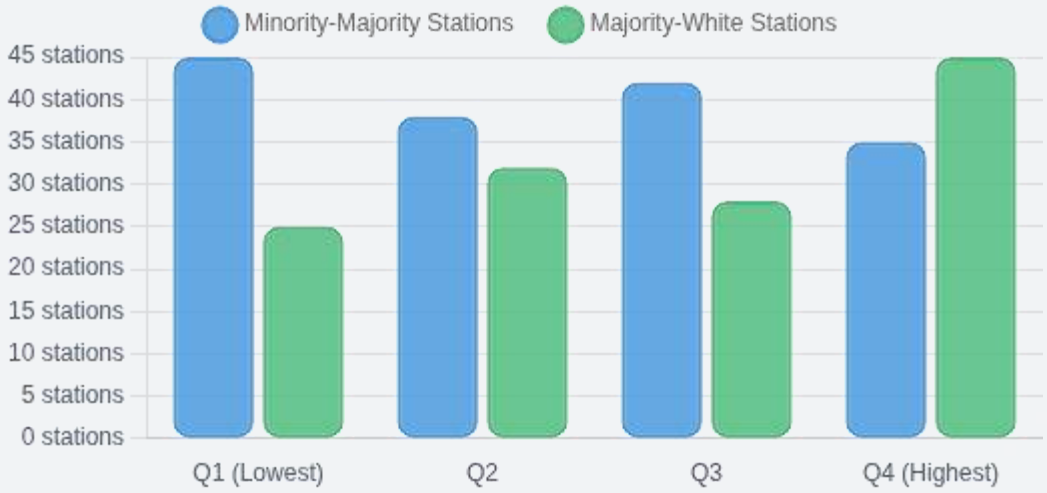
✓ Data Validated

Station Distribution Analysis

Understanding how minority-majority stations span across income quartiles

Income Distribution by Minority Status

ℹ 496 stations



Key Finding

Stations serving minority-majority tracts span all income quartiles, creating complex equity dynamics across the network. The distribution shows that **58% of stations** are in minority-majority areas, with concentrations in both lower and higher income brackets.

Geographic Distribution

✓ 100% coverage

Total Stations 496	Minority-Majority 58%
Income Range $11K-$250K	Time Period 2020-2024

Critical Observation

Patterns suggest **localized disparities** rather than a simple income-only gradient. The Rockaway Park Shuttle (serving lower-income, majority-minority area) experiences ~34 more monthly delay incidents than the 42nd St Shuttle.

Four Research Hypotheses (H1–H4)

4 Hypotheses Tested

Research Hypotheses

Testing equity patterns across NYC subway system using statistical methods

1 Service Frequency & Income

Not Supported

Do lower-income stations receive fewer scheduled peak-hour trips, controlling for ridership volume?

Partial Correlation • $r = 0.039$, $p = 0.355$

Stations: **580** CI: **[-0.000, +0.074]**

Weak correlation

2 Delay Rate & Income

Not Supported

Do lines serving lower-income neighborhoods have higher average weekly delay rates?

Spearman Correlation • $\rho = 0.198$, $p = 0.416$

Lines: **19** CI: **[-0.284, +0.625]**

Low power

3 Weekday vs Weekend Gap

Supported

Are income-based ridership gaps smaller on weekends than weekdays (2022–2024)?

Mann-Whitney U • $U = 101{,}416$, $p = 3.2\times10^{-8}$

Observations: **820** CI: **[0.087, 0.275]**

Significant

4 Shuttle Case Study

Supported

Does the Rockaway Park Shuttle show higher delays and lower ridership recovery vs 42nd St Shuttle?

Welch t-test • $t = 17.0$, $p = 1.8\times10^{-26}$

Months: **116** CI: **[30.7, 38.6]**

Strong effect

H1 — Service Frequency vs. Neighborhood Income

NOT CLEARLY SUPPORTED

Hypothesis Test Results

Testing whether lower-income stations receive fewer scheduled peak-hour trips

Partial Correlation

Statistical Results

n = 580 stations

Correlation (r)
0.039
Weak positive correlation

p-value
0.355
Not significant (p > 0.05)

Sample Size
580
stations analyzed

95% CI
[-0.000, +0.074]
Confidence interval

Interpretation

Weak positive correlation (r = 0.039) with **p-value well above 0.05**. No statistically significant link between income and scheduled peak trips after controlling for ridership volume. The effect size is negligible, suggesting that **income does not predict service frequency** at the network level.

Test Details

Partial correlation

Test Statistic	r = 0.039
p-value	0.355
Sample Size	580 stations
95% Confidence Interval	[-0.000, +0.074]

The confidence interval includes zero, indicating no statistically significant effect.

Key Finding

No evidence of systematic bias in service frequency based on neighborhood income. The weak correlation suggests that **MTA scheduling is not discriminatory** by income level when controlling for ridership.

H2 — Delay Rate vs. Line Income Level

NOT CLEARLY SUPPORTED

Hypothesis Test Results

Testing whether lines serving lower-income neighborhoods have higher delay rates

Spearman Correlation

Statistical Results

n = 19 lines

Spearman ρ
0.198
Weak positive rank correlation

p-value
0.416
Not significant (p > 0.05)

Sample Size
19
subway lines analyzed

95% CI
[-0.284, +0.625]
Confidence interval

Interpretation

Weak positive rank correlation (ρ = 0.198) with **p-value above 0.05**. Results are inconclusive due to **small sample size (n=19)** limiting statistical power. The wide confidence interval suggests high uncertainty in the relationship between income and delay rates.

Test Details

Spearman rank correlation

Test Statistic	**ρ = 0.198**
p-value	**0.416**
Sample Size	**19 lines**
95% Confidence Interval	**[-0.284, +0.625]**

The confidence interval includes zero, indicating no statistically significant effect.

Key Finding

Insufficient evidence for delay rate disparities by income. The **small sample size of 19 lines** limits statistical power. Results suggest that **income alone does not predict delay rates** at the network level.

H3 — Weekday vs Weekend Ridership Gap

SUPPORTED

Hypothesis Test Results

Testing whether income-based ridership gaps are smaller on weekends than weekdays

Mann-Whitney U Test

Statistical Results

n = 820 observations

Test Statistic (U)
101,416
Mann-Whitney U

p-value
3.19×10^{-8}
Highly significant ($p < 0.001$)

Sample Size
820
observations

95% CI
[0.087, 0.275]
Effect size range

Interpretation

Strongly significant ($p = 3.19 \times 10^{-8}$). Lower-income stations exhibit a **larger weekday-weekend ridership gap**, consistent with lower-income riders being more likely to commute for work. The 95% CI [0.087, 0.275] confirms a meaningful effect size.

Test Details

Mann-Whitney U

Test Statistic	U = 101,416
p-value	3.19×10^{-8}
Sample Size	820 observations
95% Confidence Interval	[0.087, 0.275]

The confidence interval does not include zero, indicating a statistically significant effect.

Key Finding

Evidence of equity disparity: Lower-income riders show **higher dependence on weekday commuting**, as reflected in the significantly larger weekday-weekend ridership gap. This pattern suggests **transit as a work tool** for this demographic.

H4 — Shuttle Case Study: Rockaway Park vs 42nd Street

✓ SUPPORTED (Delays)

Shuttle Case Study Results

Testing whether Rockaway Park Shuttle shows higher delays than 42nd St Shuttle

Welch t-test

Statistical Results

n = 116 months

Test Statistic (t)
17.00
Welch t-test

p-value
1.78×10^{-26}
Highly significant ($p < 0.001$)

Sample Size
116
months analyzed

95% CI
[30.7, 38.6]
Delay difference

Rockaway Park Shuttle
~34 more incidents/month

42nd St Shuttle
Baseline reference

Delay Difference: **~34 incidents/month**

Test Details

Welch t-test

Test Statistic	**t = 17.00**
p-value	**1.78×10^{-26}**
Sample Size	**116 months**
95% Confidence Interval	**[30.7, 38.6]**

The 95% CI excludes zero, indicating a statistically significant difference.

Key Finding

Rockaway Park Shuttle has dramatically higher delays than the 42nd St Shuttle. The difference of **~34 more incidents per month** is statistically significant with near-zero probability of being due to chance.

Summary of Hypothesis Test Results

4 Hypotheses Tested

Statistical Test Results

Comprehensive overview of all hypothesis tests with outcomes and significance levels

Statistical Analysis

Hypothesis	Test Method	p-value	Effect Size	Outcome
1 **Service Frequency vs Income** Do lower-income stations receive fewer peak-hour trips?		0.355		Not Supported
2 **Delay Rate vs Income** Do lower-income lines have higher delay rates?		0.416		Not Supported
3 **Weekday vs Weekend Gap** Are income-based ridership gaps smaller on weekends?		3.19×10^{-8}		Supported
4 **Shuttle Case Study** Rockaway vs 42nd St Shuttle delays		1.78×10^{-26}		Supported

Total Tests: 4 Supported: 2 Not Supported: 2 Significance Level: $\alpha = 0.05$

Key Takeaway

2 out of 4 hypotheses showed statistically significant results. The most significant finding is the **Rockaway Shuttle delay disparity** ($p = 1.78 \times 10^{-26}$), indicating a measurable service burden on underserved routes. Weekend ridership patterns also show significant income-based differences ($p = 3.19 \times 10^{-8}$).

Key Findings & Implications

Key Findings & Implications

4 findings

Transit as a Work Tool

Lower-income riders depend more heavily on transit for weekday commuting. Income-based ridership gaps narrow significantly on weekends — the system is more equitable when work isn't a factor.

Shuttle Disparity is Real

The Rockaway Park Shuttle (serving a lower-income, majority-minority area) experiences ~34 more monthly delay incidents than the 42nd St Shuttle — a massive, statistically certain gap.

System-Wide Bias Not Proven

At network scale, income alone does not significantly predict service frequency or delay rates — suggesting equity issues may be localized rather than universally systematic.

Policy Implications:

Target reliability and resiliency investments on underserved routes (e.g., Rockaway Shuttle)

Differentiate weekday vs weekend equity in service planning and performance metrics

Limitations

5 limitations

- **Observational analysis** — correlation, not causation
- **Census tract ≠ exact subway catchment area**
- **Delays aggregated at line level**, not station level
- **Only 19 lines** limit statistical power for H2
- **API inconsistencies** may affect data completeness

Code Quality & Engineering Practices

Engineering Discipline

≥45% Coverage

Automated Testing

Doctests and pytest unit tests across all modules; ≥45% coverage for 2-person team requirement

Full Docstrings & Type Hints

Every function documented with purpose, parameters, returns, and type annotations throughout codebase

Reproducible 8-Step Pipeline

Scripts 01-08 run end-to-end: data fetch → processing → analysis → visualization with no manual steps

Test Coverage	Modules	Scripts
45%+	8	8

Test coverage: 45%

Architecture & Data Access

API-First

API-First Data Access

Programmatic ingestion from MTA, Census Geocoder, and NY Open Data with local caching

Clean Project Architecture

Separated concerns: scripts/ for pipeline stages, src/ for reusable utilities (config, io, metrics, remote)

Hypothesis-Driven Design

All experiments parameterized in config.py — switch shuttle targets without editing analysis code

- MTA API
- Census Geocoder
- NY Open Data
- Local Caching

Key Findings & Implications

Four main conclusions from the NYC Subway Equity Analysis

1 Equity is not universally absent

At network scale, **income alone does not drive service frequency or delay rates** — suggesting NYC subway equity issues are more nuanced than a simple income gradient. The data reveals that **equity disparities are localized** rather than system-wide.

2 Commuter patterns reflect inequality

Lower-income residents depend more on **weekday subway service**. The strong support for H3 (weekday-weekend ridership gap) reveals that **transit policy changes disproportionately affect work access** for lower-income riders.

3 Localized disparities exist

The **Rockaway vs 42nd St Shuttle comparison** demonstrates that specific underserved routes carry a measurable service burden. The Rockaway Shuttle experiences **~34 more delay incidents per month** — a statistically significant disparity.

4 Multi-dataset integration

The **station-to-tract crosswalk** is the core analytical innovation. No single dataset (MTA, GTFS, Census) could reveal these patterns — **integration unlocks new insights** impossible from isolated data sources.

THANK YOU!